

SQL Joins Assignment (50 Questions)

Practice Tables

1) students

```
CREATE TABLE students (  
  student_id INT PRIMARY KEY,  
  student_name VARCHAR(50),  
  city VARCHAR(30)  
);
```

```
INSERT INTO students VALUES
```

```
(1, 'Amit Sharma', 'Delhi'),  
(2, 'Priya Verma', 'Mumbai'),  
(3, 'Ravi Kumar', 'Pune'),  
(4, 'Neha Singh', 'Delhi'),  
(5, 'Rahul Mehta', 'Bangalore'),  
(6, 'Kiran Joshi', 'Hyderabad');
```

2) courses

```
CREATE TABLE courses (  
  course_id INT PRIMARY KEY,  
  course_name VARCHAR(50),  
  duration_months INT  
);
```

```
INSERT INTO courses VALUES
```

```
(101, 'SQL Basics', 2),  
(102, 'Python Programming', 3),  
(103, 'Data Science', 4),  
(104, 'Web Development', 5);
```

3) enrollments

```
CREATE TABLE enrollments (  
  enrollment_id INT PRIMARY KEY,  
  student_id INT,  
  course_id INT,  
  enrollment_date DATE  
);
```

```
INSERT INTO enrollments VALUES
```

```
(1001, 1, 101, '2025-06-01'),  
(1002, 2, 102, '2025-06-02'),  
(1003, 3, 103, '2025-06-03'),  
(1004, 1, 102, '2025-06-04'),  
(1005, 4, 104, '2025-06-05'),  
(1006, 3, 101, '2025-06-06');
```

Assignment: 50 Questions on Joins

INNER JOIN (15 Questions)

1. Retrieve all students with their enrolled course names.
2. Retrieve student names, course names, and enrollment dates.
3. List student names and course names where the course duration is more than 2 months.
4. Retrieve all enrollments with student city.
5. Retrieve all enrollments with course duration.
6. List student names enrolled in 'SQL Basics'.
7. Retrieve course names for students living in 'Delhi'.
8. Find students enrolled in 'Data Science' with their cities.
9. Retrieve student names, course names, and enrollment IDs.
10. List student names with enrollment dates where course duration is less than 5 months.
11. Retrieve student and course details for enrollment IDs greater than 1003.
12. Retrieve all students with their courses where enrollment date is after '2025-06-02'.
13. Retrieve the student name and course name for student ID 1.
14. List the course name and student name for courses with a duration of 3 months.
15. Retrieve enrollment details with student names for students living in 'Bangalore'.

LEFT JOIN (15 Questions)

16. Retrieve all students with their enrolled courses, including students not enrolled in any course.
17. List student names, course names, and enrollment dates for all students, including those without enrollments.
18. Find all students and their courses if enrolled; if not, show NULL for course.
19. List all students with their course names if they are enrolled in a course with a duration of 4 months.
20. Retrieve all students and their enrollment dates if enrolled; if not, show NULL.
21. List student names with their enrollment dates for all students.
22. Find students living in 'Delhi' and their courses if enrolled.
23. Retrieve all students with course details where available.
24. Find all students and courses where course duration is 5 months.
25. Retrieve student names with course names for student IDs greater than 2.
26. List all students and their enrolled course names for students living in 'Pune'.
27. Retrieve all students with enrollment IDs if they are enrolled.
28. List all students with courses they are enrolled in, even if the course name is 'Web Development' and the student is not enrolled.
29. Retrieve all students with enrollment details for enrollment dates in June.
30. List all students and courses they are enrolled in, including students without enrollment.

RIGHT JOIN (10 Questions)

31. Retrieve all courses with enrolled student names, including courses with no enrolled students.
32. List course names with student names for all courses.
33. Retrieve all courses with student details if enrolled.
34. List courses and enrolled student names where course duration is 3 months.
35. Retrieve all courses with enrollment dates of students.
36. Find courses and student names where course ID is 101.
37. Retrieve course names and student names for enrollments made in June.
38. List courses and student names for courses with a duration of 5 months.
39. Retrieve all courses and student names for students living in 'Mumbai'.
40. Find all courses with their enrolled students or NULL if no student is enrolled.

FULL JOIN (Simulated) (10 Questions)

41. Retrieve all students and their enrolled courses, including students without enrollments and courses without students.
42. List all students and course names using FULL JOIN simulation.
43. Retrieve all students and course details, including unmatched data.
44. Retrieve all enrollments, student names, and course names including unmatched students and courses.

45. List student names with course names for all courses and students.
46. Retrieve all students with courses enrolled and courses without any students using FULL JOIN.
47. List all courses with enrolled students, including students without courses.
48. Retrieve all students and courses, showing NULL where there is no match.
49. Find all students and courses using FULL JOIN simulation for reporting.
50. Retrieve all students and courses, including students without enrollments and courses without enrollments.